

THE UNIVERSITY OF  
**SYDNEY**

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|----------------|--|
| Tutorial Class |  |
| Student Name   |  |
| SID            |  |

**CONFIDENTIAL QUIZ PAPER**

This paper should not be removed from the quiz venue.

**COMP5339 – Data Engineering**

Tutorial Quiz – 1

Individual Quiz (5%)

Quiz Writing Time: **30 minutes**  
**Friday, 22 May, at 08 PM**

**Instructions to Students**

- Please ensure you correctly complete your **tutorial** and **Student ID** details.
- This is a **closed-book** test. **Electronic devices** (including phones, laptops, tablets, etc.) are **not permitted**. Please place all devices in your bag and keep your bag on the floor.
- This tutorial quiz consists of **3 multiple-choice questions** and three short descriptive questions. **All questions must be answered.**
- Write all answers in the spaces provided on this paper.
- Questions carry **different marks**. The mark value for each question is shown at the start of the question. The **total mark** for this quiz is **10**, which contributes to 5% of the total marks.
- For short-answer questions, please write **legibly**. Record your final answers in **ink**. **Do not** use a pencil or **red ink**.

Please submit the completed quiz paper to your tutor **before leaving the room**.

## Part I – Multiple Choice Questions

**Question 1:** This question has three parts. Tick the boxes corresponding to each correct answer. Each question carries one mark.

**A.** Which of the following is an example of advanced feature engineering for time-series data?

- ☐ Stop-word removal
- ☐ GPS metadata extraction
- ☐ Binary thresholding only
- ☐ **Lag features and rolling averages**

**B.** In the ML pipelines, what is the strongest reason to keep preprocessing and feature extraction consistent between training and inference?

- ☐ **To ensure the model receives inputs in the same feature space it learned from**
- ☐ To make slides look cleaner
- ☐ To avoid using storage systems
- ☐ To convert all unstructured data into metadata

**C.** In a DSMS, queries are typically:

- ☐ One-time and transient
- ☐ **Continuous and persistent**
- ☐ Only executed at the end of the day
- ☐ Limited to batch tables

## Part II – Short Answer Questions

Provide answers in the provided boxes.

**Question 1:** What is a session window? Provide one challenge in a streaming pipeline. (2 Marks)

A session window is a variable-length window that groups related events based on activity boundaries, such as explicit markers or inactivity gaps. It is useful for modelling sessions such as auctions or user activity periods.

**Question 2:** What is meant by scale-agnostic data management and scale-agnostic data processing? Explain how sharding, replication, and parallel processing contribute to these goals. (2 Marks)

Scale-agnostic data management means storing and organising data so the system can grow smoothly, using techniques such as sharding for performance and replication for availability. Scale-agnostic data processing means computation can be parallelised across many CPUs or nodes. Parallel processing improves performance, while sharding and replication help the system scale and remain available under failures.

**Scenario 3:** A company currently uses an ETL-based data warehouse and wants to migrate to an ELT-based cloud warehouse. What major architectural changes would be required, and why might this migration be difficult from a data engineering perspective? **(3 Mark)**

The company would need to introduce or expand a staging/raw layer, move transformation logic closer to the warehouse, redesign schemas and orchestration workflows, and update governance, validation, and monitoring processes. The migration is difficult because it affects the core data flow, existing dependencies, reporting logic, team responsibilities, and infrastructure design.

EXTRA SHEET